

Nmap Command Reference & Cheat Sheet

Essential Network Discovery & Vulnerability Scanning Guide for Security Analysts

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1. BASIC TARGET SPECIFICATION

nmap 192.168.1.1	Scan a single IP address
nmap target.com	Scan a domain name
nmap 192.168.1.1-20	Scan a range of IP addresses
nmap 192.168.1.0/24	Scan an entire CIDR subnet
nmap -iL targets.txt	Scan targets from a file list

2. PORT SELECTION TECHNIQUES

nmap -p 80 target	Scan specific port (HTTP)
nmap -p 22,80,443 target	Scan multiple specific ports
nmap -p 1-1024 target	Scan port range (1 to 1024)
nmap -p- target	Scan ALL 65,535 ports
nmap -F target	Fast scan (100 most common ports)
nmap --top-ports 20 target	Scan top 20 most frequent ports

3. SCAN TYPES & MECHANICS

nmap -sS target	SYN Stealth Scan (Default / Fast / Stealthy)
nmap -sT target	TCP Connect Scan (Full handshake)
nmap -sU target	UDP Scan (DNS, DHCP, SNMP)
nmap -sA target	ACK Scan (Map firewall rulesets)
nmap -sN / -sF / -sX	NULL, FIN, and Xmas Scans

4. SERVICE & OS DETECTION

nmap -sV target	Detect service versions running on ports
nmap -O target	Identify remote Operating System
nmap -A target	Aggressive scan (OS, Service, Scripts, Traceroute)
nmap -sV --version-intensity 5	Set version probe intensity (0-9)

5. NMAP SCRIPTING ENGINE (NSE)

nmap -sC target	Run default safe Nmap scripts
nmap --script=vuln target	Run vulnerability detection scripts
nmap --script=http-title target	Run a specific individual script
nmap --script "http-*" target	Run all HTTP-related scripts

6. OUTPUT & PERFORMANCE OPTIONS

nmap -oN scan.txt target	Normal output format
nmap -oX scan.xml target	XML output format
nmap -oA result target	Output in all 3 major formats
nmap -T4 target	Aggressive timing template (Fast)
nmap -Pn target	Treat all hosts as online (Skip ping)

7. PRO TIP - COMPREHENSIVE RECONNAISSANCE COMMAND

```
nmap -sC -sV -p- -T4 -oA initial_scan <TARGET-IP>
```